

Juan Pablo de Legarreta

NETC 170 – IT Governance & Management

Module Eight Assignment

Cybersecurity KPI

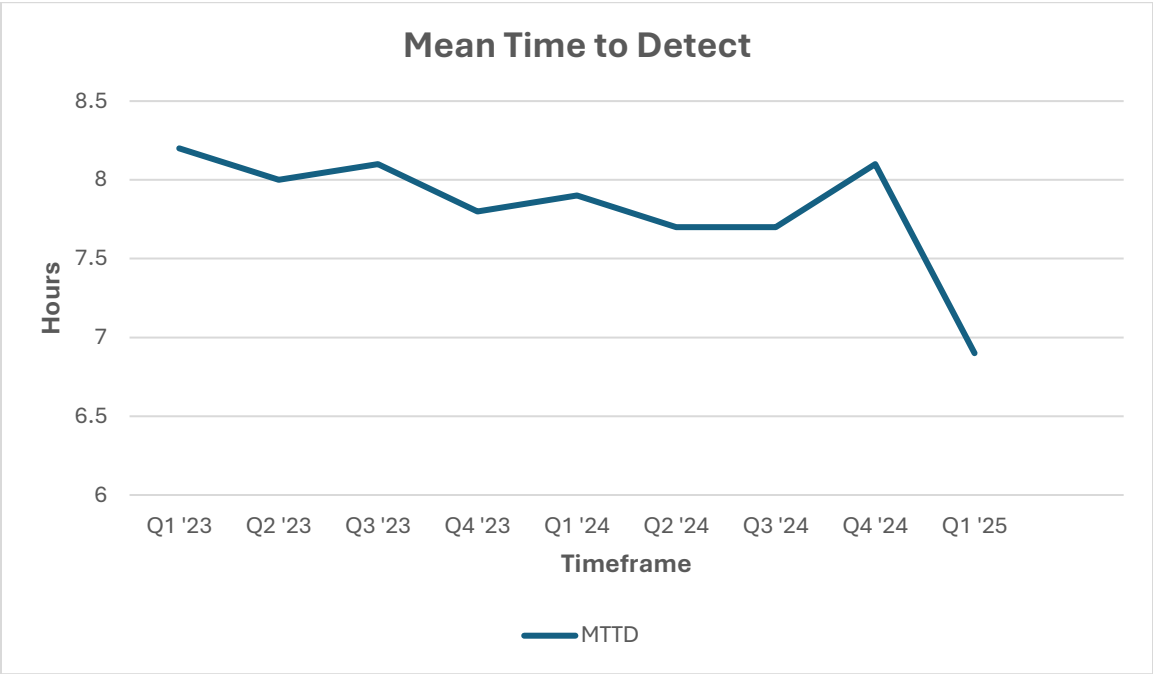
Key Performance Indicators (KPIs) measure “performance around strategic business goals.” (Bonnie) An important cybersecurity KPI is Mean Time to Detect or MTTD. MTTD measures “the average amount of time between when an incident occurs and when it’s detected.” (Bonnie) Because of this, MTTD can be a great barometer for how well anti virus/malware software is working, how well an incident detection team is trained, and how effective the organization’s measures are for incident detection. MTTD can also be a starting point to calculate average cost per security incident and also an organization’s return on security investment.

For this assignment, I would like to use fictional data about MTTD to tell a story justifying an expensive onboarding training on for the recently expanded security team. In my fictional scenario, this data will be used in a quarterly board meeting to justify the hiring of 4 new employees, showing how their addition has been very valuable to the security team and to the organization overall. In order to accomplish this, I will be plotting these fictional data points on a line graph – the best graph to show change over time – demonstrating the downward trend of mean time to detect security events. (John James) This will demonstrate how increasing the size of the team has decreased the mean time to detect and thus has had a good return on security investment.

The Data and Charts

My fictional data points represent the Mean Time to Detect in hours in every quarter since the beginning of 2023. Below you will find a line graph and a table with values of the

MTTD in hours over every quarter from Q1 of 2023 until Q2 of 2025. Over the two and a half year span, you can see the trends of the MTTD.



Timeframe	MTTD (hours)	Timeframe	MTTD (hours)	Timeframe	MTTD (hours)
Q1 2023	8.2	Q1 2024	7.9	Q1 2025	6.9
Q2 2023	8	Q2 2024	7.7		
Q3 2023	8.1	Q3 2024	7.7		
Q4 2023	7.8	Q4 2024	8.1		

The figures reflect a downward trend – that for a couple of quarters (Q1-Q3 of 2023) the rate stayed about the same, hovering around 8 hours, which steadily declines to a plateau at 7.7 hours between Q2 and Q3 of 2024. Then there is a spike in Q4 of 2024. This is when the four new employees were hired. Understandably, MTTD goes back up as they transition into the role. However, as they get used to it, MTTD trends sharply downward, displaying the benefit of adding the new employees – in just a couple of months they were able to decrease MTTD.

Conclusion

The data on the graph confirms the assertion that the new staff members have created a downward trend in MTDD. For this reason, the graph supports the story I wanted to tell. Though there is a temporary bump upward as the new staff got hired, it clearly dips (at a higher rate than before) once they became more comfortable with their jobs. Decreased MTDD decreases overall costs related to security incidents, downtime, man-hours, and data loss. Adding 4 new employees to the security team has had an immediate positive impact that will affirm the decision to hire them.

Sources

Bonnie, Emily. "The 10 Most Important Cybersecurity Metrics & KPIs for CISOs to Track." *Secureframe*, 21 Jan. 2023, <https://secureframe.com/blog/cybersecurity-metrics-and-kpis>.

John James. "How to Use Graphs to Tell Your Story." *YouTube*, 1 Mar. 2021, <https://www.youtube.com/watch?v=ksfpEa6HSn0>. Accessed 8 Apr. 2025.